
TECHNICAL PROPOSAL FOR DEVELOPMENT OF AN EMBEDDED SOLUTION FOR REAL -TIME IMAGE ANALYSIS USING AI

AGENDA

01

Project Overview & Problem Statement

Understanding the fire threat detection challenge and embedded AI opportunity

02

Objectives & Scope

Key goals, deliverables and system boundaries

03

System Architecture Overview

Hardware-software integration design and data flow

04

AI Algorithm Pipeline

Flame/smoke signature detection model design

05

Hardware Design & Selection

Optimum embedded platform and component selection

06

Software & Firmware Design

Model deployment, real-time processing pipeline

07

Project Milestones & Team

Phased delivery plan, budget summary and proposed team

PROJECT OVERVIEW & PROBLEM STATEMENT

Current Challenges

- Traditional smoke/heat detectors have high false-alarm rates
- No visual intelligence — cannot distinguish flame signatures from steam or dust
- Slow response time in large industrial or outdoor environments
- Expensive centralized server-based AI systems not feasible for field deployment
- No real-time edge processing capability in existing low-cost systems
- Lack of self-learning/adaptable models in deployed fire detectors

The Opportunity

- AI models can visually detect flame & smoke signatures with high accuracy
- Edge AI (on-device) enables real-time detection with no cloud dependency
- Lightweight models (YOLO-Nano, MobileNet) can run on affordable embedded boards
- Integrated alarm & control system possible within single embedded unit
- Low power consumption enables battery/solar-powered deployments
- Scalable prototype for industrial, residential and defense applications

OBJECTIVES & SCOPE



Develop AI Model

Train/deploy a lightweight deep learning model capable of detecting flame and smoke signatures in real-time video frames with >90% accuracy



Embedded Deployment

Port and optimize the AI model to run efficiently on an embedded platform (e.g., NVIDIA Jetson Nano) with minimal latency



Real-Time Image Processing

Build a continuous video capture and inference pipeline processing live camera frames at ≥ 15 FPS



Alarm & Control Integration

Interface the detection output with hardware alarm systems (buzzer, LED, relay) for automated fire response



Optimum Resource Usage

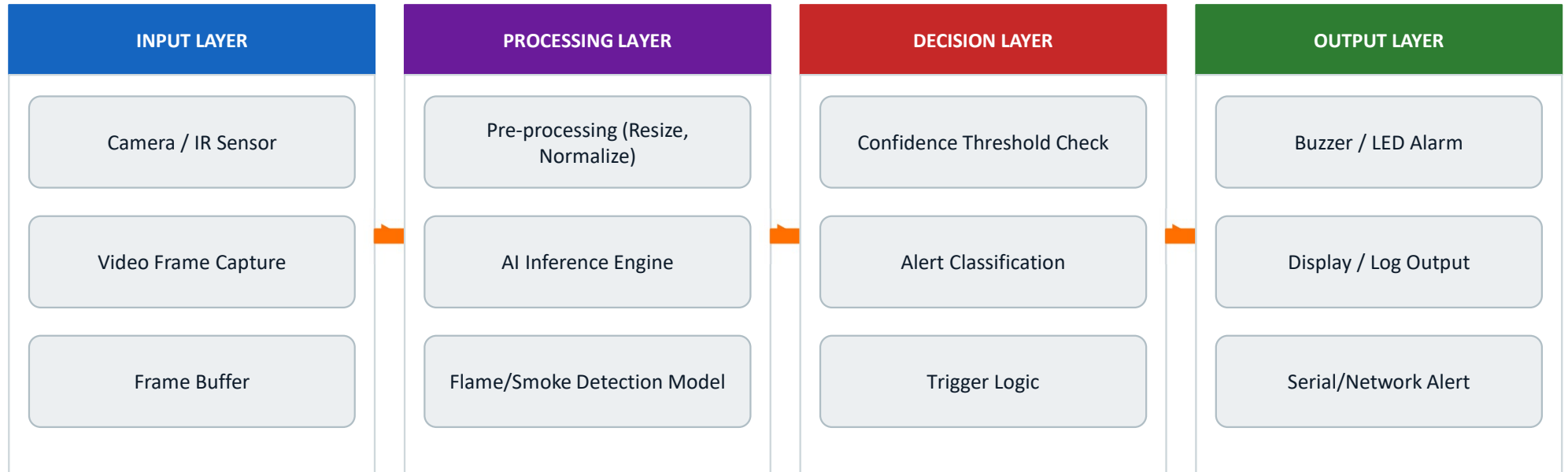
Ensure CPU/GPU usage, memory footprint and power consumption remain within embedded hardware limits



Prototype for Fire Control System

Deliver a working prototype ready for integration into a larger fire detection & control infrastructure

SYSTEM ARCHITECTURE OVERVIEW



AI ALGORITHM PIPELINE

01 Data Collection & Dataset

- Fire/smoke image datasets (e.g., VisiFire, CAIR)
- Augmentation: flip, rotate, brightness shift
- Train/Val/Test split: 70/20/10

02 Model Architecture

- YOLOv8-Nano / MobileNet-SSD
- Custom CNN trained on flame+smoke classes
- Transfer learning from COCO pretrained weights

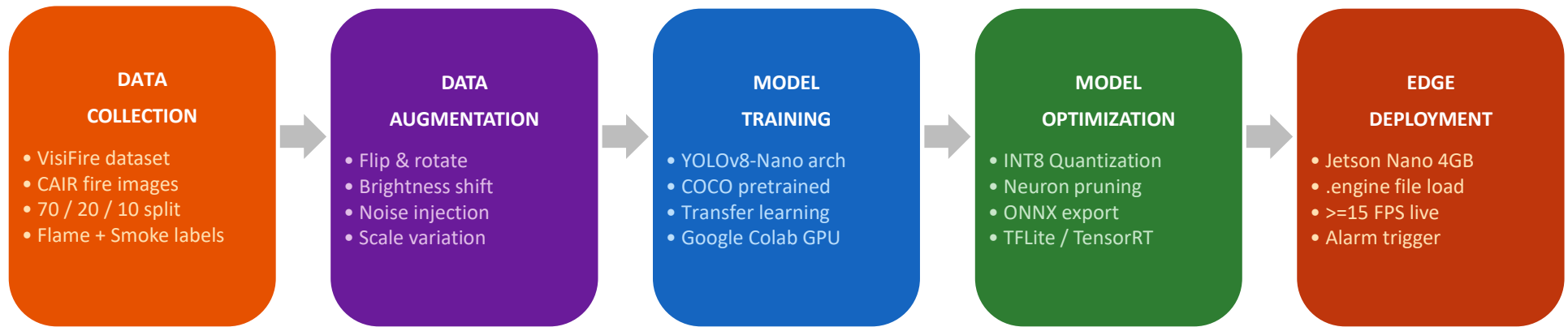
03 Model Optimization

- Quantization: FP32 → INT8 (4x size reduction)
- Pruning of low-weight neurons
- TensorRT / ONNX / TFLite export for edge

04 Inference & Post-Processing

- Real-time frame inference at ≥ 15 FPS
- Non-Maximum Suppression (NMS) for bounding boxes
- Confidence threshold: 0.6 (tunable)

AI MODEL TRAINING & OPTIMIZATION PIPELINE



HARDWARE DESIGN & SELECTION



Processing Unit

- NVIDIA Jetson Nano Developer Kit 4GB — GPU-accelerated option



Imaging / Sensing

- IMX219-160 Camera (Jetson-compatible CSI)



Alarm & Output

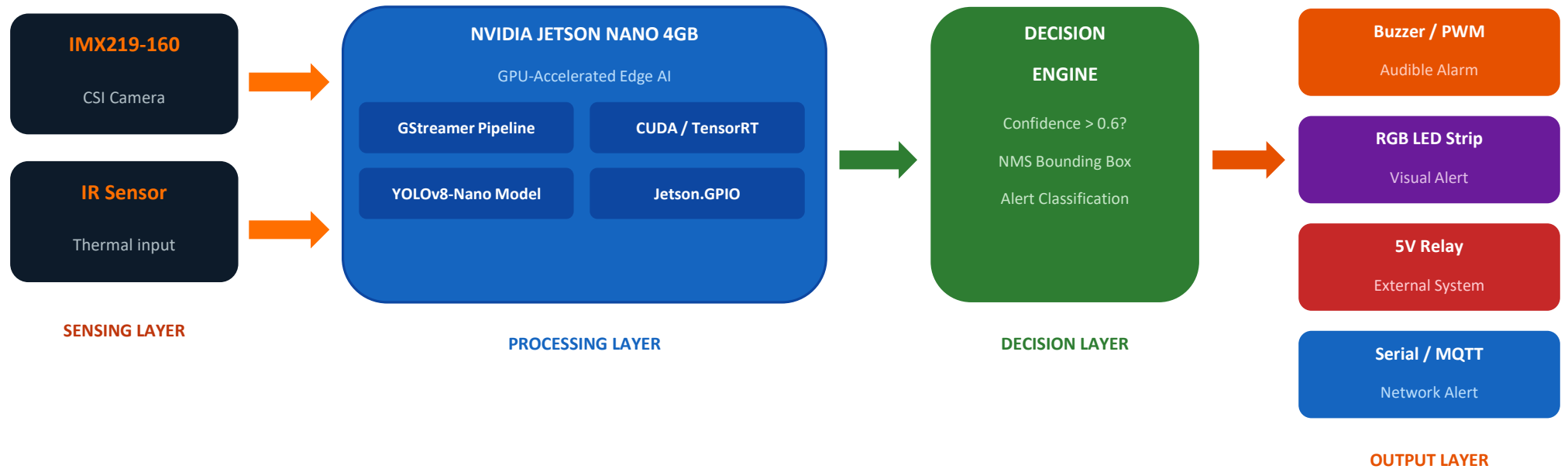
- Piezo Buzzer (PWM-controlled)
- RGB LED strip for visual alert
- 5V Relay module for external systems



Power & Connectivity

- 5V/4A power supply (Jetson)
- MicroSD 64GB (model storage + OS) or SSD

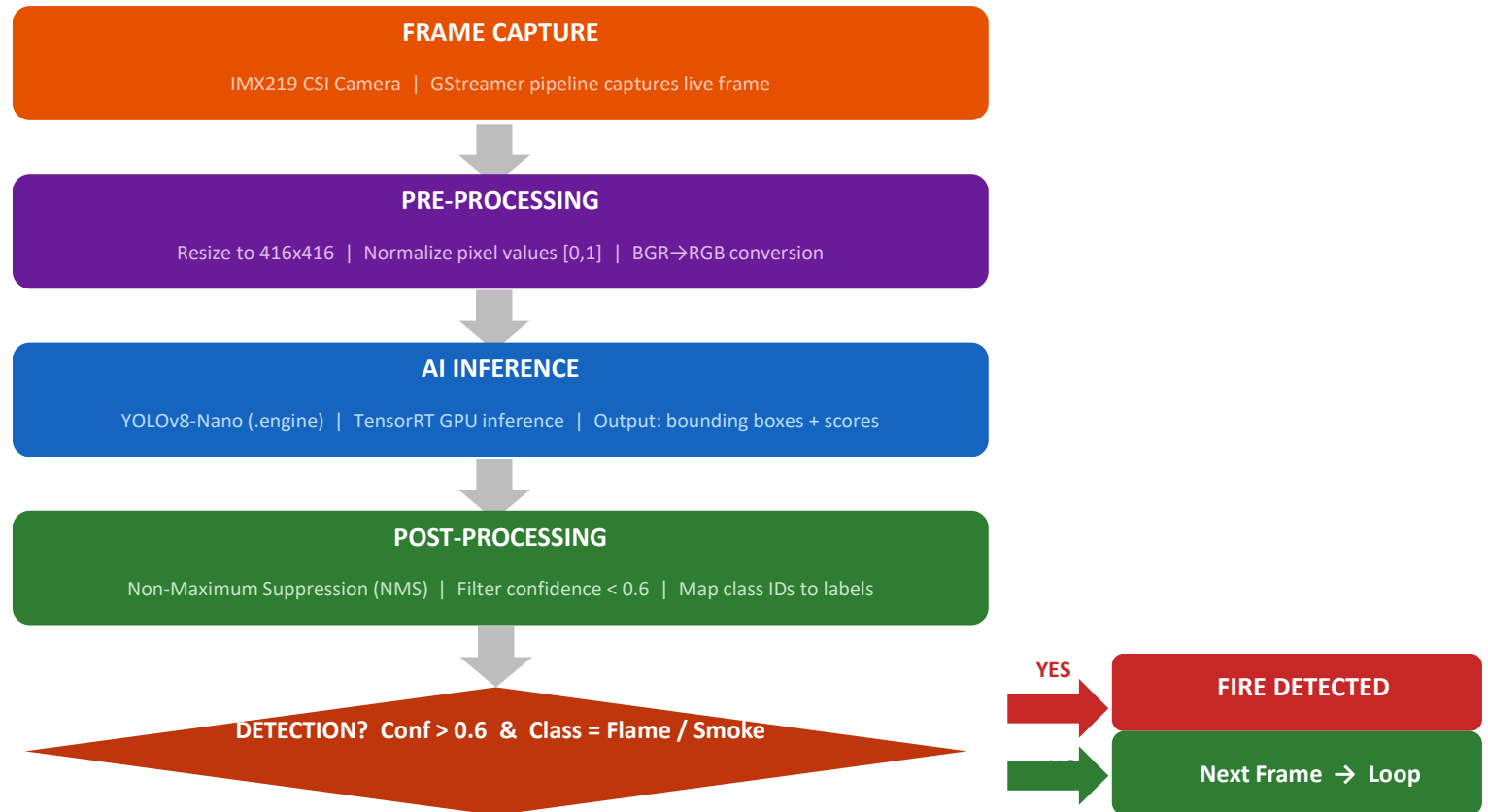
HARDWARE INTEGRATION BLOCK DIAGRAM



SOFTWARE & FIRMWARE DESIGN

Layer	Tool / Library	Version
OS / Runtime	NVIDIA JetPack SDK	4.6.x
AI Framework	YOLOv8 + TensorRT	YOLOv8.2 / TRT 8.x
Camera Pipeline	Gstreamer + nvarguscamerasrc	Built-in
Computer Vision	OpenCV (CUDA-enabled)	4.5.x
GPIO Control	Jetson.GPIO	Latest
Communication	mqtt	1.6.x
Logging	SQLite3 + Python	Built-in
Model Training	Ultralytics + Google Colab	YOLOv8.2
Model Export	ONNX → TensorRT (.engine)	On-device
Language	Python 3.8	JetPack default

REAL-TIME INFERENCE PIPELINE



PROJECT MILESTONES & TIMELINE

Phase 1 Week 1–2	Literature Review & Dataset Prep Survey existing fire detection models, Collect & label fire/smoke datasets, Set up embedded hardware
Phase 2 Week 3–6	AI Model Development Train baseline CNN/YOLO model, Evaluate on test dataset, Optimize with quantization & pruning
Phase 3 Week 7–8	Embedded Deployment Port model to TFLite/OpenVINO, Integrate with camera pipeline, Achieve real-time ≥ 15 FPS inference
Phase 4 Week 9–11	Alarm & Control Integration Wire GPIO alarms (buzzer, LED, relay), Test end-to-end fire detection $\rightarrow$ alarm flow
Phase 5 Week 12–14	System Testing Real-world fire/smoke testing, False positive reduction tuning, latency + FPS optimization
Phase 6 Week 15-16	Final Prototype, Demo & Documentation Full system testing & accuracy validation, Prepare final demo prototype & report

EXPERIENCE IN AI AND MACHINE LEARNING EXPERTISE

- Indian Army
 - AI based Electro- Surveillance System
- Troop Comfort
 - Future Soldier System (GIS, NLP, Prediction)
- Indian Navy-WESEE
 - Maritime Awareness Platform,(GIS, LLM, Predictions, LLM, NLP)
 - Agentic AI based Infrastructure Upgrade Detection System
- DRDO ADRDE
 - Net Height Measurement System
- Indian Army - DGIS (AI incubation Center)
 - Sensor to Shoot
- IIITA
 - AI Based Covid Awareness App , uses NLP Vernacular languages

SIMILAR EXPERIENCE IN AI

➤ Indian Army, Medium Range, Deep Shooters, Mathura : AI based Electro Surveillance Software

1. The software include integration with various sensors to get video Streaming and other feedback. The System use AI/ML to detect class of Object and calculate threats and raise alerts.
2. The System is capable of Self Learning on Models which were previously identified or classified.

➤ Indian Institute of information Technology, Allahabad : AI /NLB based App for COVID-19 Awareness

1. The System aims to help people through spreading awareness about Covid – 19 disease or any other disease which may outbreak in future. It will broadcast all the relevant information on the disease .
2. It will also give the Dos and Don'ts for prevention and cure of disease.
3. It helps people to access medical help in vernacular language.
4. System through Artificial Intelligence and Natural Language Processing can identify the symptom /disease and can prescribe preliminary medicines or can guide the patient for doctor consultation.

➤ Ariel Delivery and Research Development (ADRDE- DRDO) : Net Height Measurement Software

1. Software uses image Processing , detection , pattern recognition and Mathematical calculations to calculate the height of Safety Net

➤ Instruments Research And Development Establishment,- DRDO Dehradun : High Frame Rate Processing algorithm implementation(AI based Image processing)

SIMILAR EXPERIENCE IN AI

➤ **Maritime Awareness Platform : Situational Awareness Platform , Indian Navy , New Delhi**

1. Comprehensive Maritime Picture: The platform provides a real-time, integrated view of all maritime activities from different data sources within a region, giving common operating pictures.
2. AI Based Features
3. Early Threat Detection, Anomaly detection. Risk Assessment , Threat Prediction, Incident Detection & Response , Predictive Analysis , Decision Support
4. Integration with Data Warehouse and Data Lake for Large scale Data Collection and processing,

PROPOSED TEAM

- Project Oversight : Avneesh Gupta
- Industry Expert (Technical) : Arvind Kumar Agrawal- Ex. Sup Engineer (Executive Shipping and MSC)
- Industry Expert (Navigation) : Cmde(Retired) Ajay Kumar Agrawal
- Sharanya Y
 - Technical Leader- AI , GIS & Defense Systems Expert
- Shantanu
 - Sr Software Engineer – AI, GIS, & Defense Systems Expert
- Venkatesh Shanigarapu-
 - Developer AI (LLM) , NLP
- Viswateja M
 - Developer – AI and GIS
- Ganesh
 - Developer – AI and Defense Systems